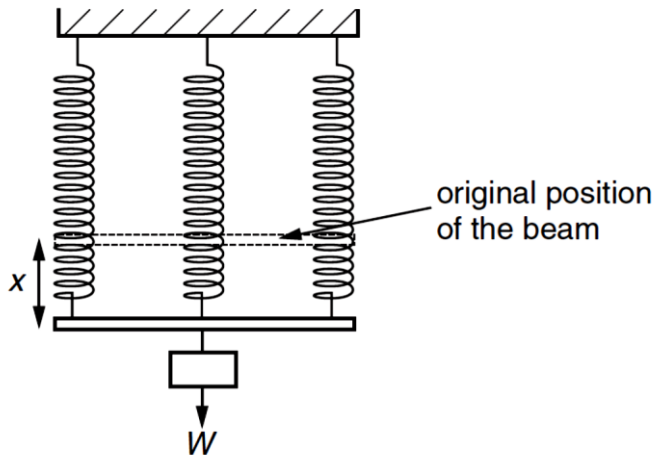


$$2u = v_2 - v_1 \quad (\text{eliminates B, C \& D})$$

$$u = 2v_1 + v_2 \quad (\text{eliminates B, C \& D})$$

- 7 A beam, the weight of which may be neglected, is supported by three identical springs. When a weight W is hung from the middle of the beam, the extension of each spring is x .



The middle spring and the weight is removed.

What is the extension when a weight of $2W$ is hung from the middle of the beam?

7

A $\frac{4x}{3}$

B $\frac{3x}{2}$

C $2x$

D $3x$

Ans: D

When W is doubled, x for each spring doubles, hence $2x$. But when one spring is removed, the new extension is thus $(2x) \times (3/2) = 3x$.

8. A gas at a pressure of $5.0 \times 10^5 \text{ Pa}$ is enclosed in a cylinder fitted with a piston.